Experiment No: 1

Date: 15/09/2026

Title: Bank Account Management System using Inheritance and Method Overriding

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Objective / Aim

To design and implement a **Bank Account Management System** in Java using **classes, objects, inheritance, constructors, methods, method overriding, deposits, withdrawals, interest calculation, and fund transfers**.

System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

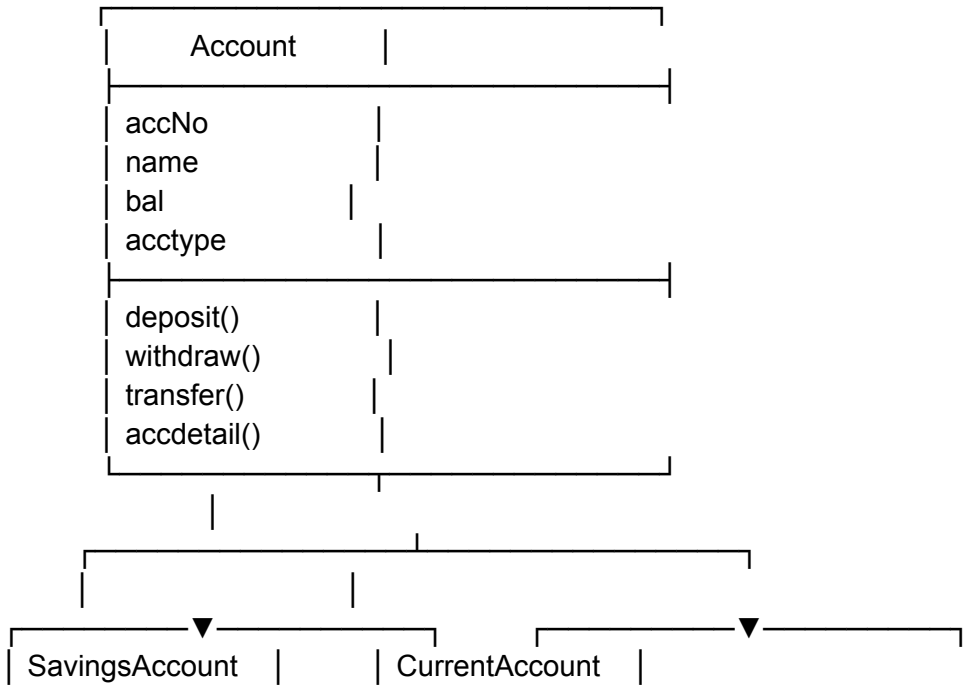
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or later)
- Text Editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans
- Command Prompt / Terminal

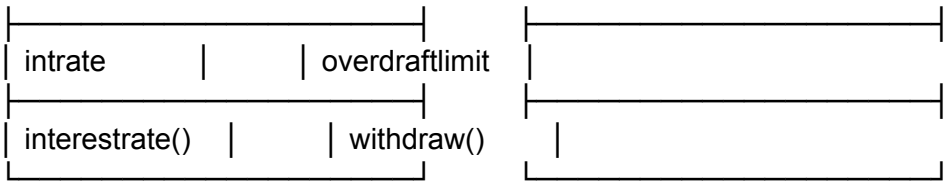
Algorithm / Logic Description

1. Create a base class named `Account`.
2. Define the following attributes:
    - Account number

    - Account holder name
    - Balance
    - Account type
3. Create a constructor in the `Account` class to initialize account details.
4. Define `deposit()` method to add money into the account.
5. Define `withdraw()` method to withdraw money if sufficient balance is available.
6. Define `transfer()` method to transfer money from one account to another.
7. Define `accdetail()` method to display account holder details.
8. Create a `SavingsAccount` class that inherits from `Account`.
9. Add an `intrate` attribute to the `SavingsAccount` class.
10. Define `interestrate()` method to calculate and add interest to the savings account.
11. Create a `CurrentAccount` class that inherits from `Account`.
12. Add an `overdraftlimit` attribute to the `CurrentAccount` class.
13. Override the `withdraw()` method in `CurrentAccount` to allow withdrawal up to the overdraft limit.
14. Create objects for `SavingsAccount` and `CurrentAccount`.
15. Perform deposits, withdrawals, interest calculation, and money transfer.
16. Display the final account details.

UML Class Diagram





Relationship: `SavingsAccount` and `CurrentAccount` inherit from the `Account` class.

`CurrentAccount` overrides the `withdraw()` method of the `Account` class.

Source Code

```
// creating a class for account
class Account {

    int accNo;
    String name;
    int bal;
    String acctype;

    // constructor
    Account(int accNo, String name, int bal, String acctype) {
        this.accNo = accNo;
        this.name = name;
        this.bal = bal;
        this.acctype = acctype;
    }

    // method to deposit money
    void deposit(int amount) {
        bal = bal + amount;

        System.out.println(
            "amount is deposited into account succesfully : " + amount
        );

        System.out.println("balance updated to " + bal);
    }

    // method to remove money
    void withdraw(int amount) {
```

```
    if (amount > bal) {
        System.out.println("balance insufficent");
    }
    else {
        bal = bal - amount;

        System.out.println(
            "withdraw sucessfully :" + amount
        );

        System.out.println(
            "balance is upated to :" + bal
        );
    }
}

// to transfer money from one account to another
void transfer(Account name, int amount) {

    if (amount > bal) {
        System.out.println("balance insufficent");
    }
    else {
        name.bal = name.bal + amount;
        bal = bal - amount;

        System.out.println("transacation done successfully");

        System.out.println(
            "balance in acc is :" + bal
        );
    }
}

// to present account holder details
void accdetail() {

    System.out.println(
        "account holder name" + name
    );

    System.out.println(
        "account no is " + accNo
```

```
        );

        System.out.println(
            "balance in account" + bal
        );

        System.out.println(
            "account type" + acctype
        );
    }
}


// creating a class for savings account
class SavingsAccount extends Account {

    double intrate;

    // constructor
    SavingsAccount(
        int accNo,
        String name,
        int bal,
        double intrate
    ) {
        super(accNo, name, bal, "savings");
        this.intrate = intrate;
    }

    // method to calculate interest rate
    void interestrate() {

        double interest = bal * intrate / 100;

        bal = bal + (int) interest;

        System.out.println(
            "interest is " + interest
        );

        System.out.println(
            "balance after interest is " + bal
        );
    }
```

```
}


// creating a class for current account
class CurrentAccount extends Account {

    int overdraftlimit;

    // constructor
    CurrentAccount(
        int accNo,
        String name,
        int bal,
        int overdraftlimit
    ) {
        super(accNo, name, bal, "current");
        this.overdraftlimit = overdraftlimit;
    }

    // overriding withdraw method
    @Override
    void withdraw(int amount) {

        if (amount > bal + overdraftlimit) {
            System.out.println(
                "withdrawal exceeds overdraft limit"
            );
        }
        else {
            bal = bal - amount;

            System.out.println(
                "withdraw suscessfully :" + amount
            );

            System.out.println(
                "balance is upated to :" + bal
            );
        }
    }
}


// main class
```

```
public class TaskII {

    public static void main(String[] args) {

        // creating savings account object
        SavingsAccount s1 =
            new SavingsAccount(
                101,
                "Naveen",
                10000,
                5
            );

        // creating current account object
        CurrentAccount c1 =
            new CurrentAccount(
                102,
                "Rahul",
                5000,
                3000
            );

        System.out.println(
            "----- SAVINGS ACCOUNT DETAILS -----"
        );

        s1.accdetail();

        System.out.println(
            "\n----- CURRENT ACCOUNT DETAILS -----"
        );

        c1.accdetail();

        System.out.println(
            "\n----- TRANSACTIONS -----"
        );

        // depositing money into savings account
        s1.deposit(2000);

        // withdrawing money from savings account
        s1.withdraw(1000);
```

```
        // calculating interest
        s1.interestrate();

        // depositing money into current account
        c1.deposit(2000);

        // withdrawing money from current account
        c1.withdraw(9000);

        // transferring money from savings to current
        s1.transfer(c1, 2000);

        System.out.println(
            "\n----- FINAL ACCOUNT DETAILS -----"
        );

        System.out.println(
            "\nSavings Account"
        );

        s1.accdetail();

        System.out.println(
            "\nCurrent Account"
        );

        c1.accdetail();
    }
}
```

Compilation & Execution Commands

Save the source code exactly as:

TaskII.java

Compile the program:

javac TaskII.java

Run the program:

java TaskII

Output Screen

```
----- SAVINGS ACCOUNT DETAILS -----
account holder nameNaveen
account no is 101
balance in account10000
account typesavings

----- CURRENT ACCOUNT DETAILS -----
account holder nameRahul
account no is 102
balance in account5000
account typecurrent

----- TRANSACTIONS -----
amount is deposited into account succesfully : 2000
balance updated to 12000

withdraw suscessfully :1000
balance is upated to :11000

interest is 550.0
balance after interest is 11550

amount is deposited into account succesfully : 2000
balance updated to 7000

withdraw suscessfully :9000
balance is upated to :-2000

transacation done successfully
balance in acc is :9550

----- FINAL ACCOUNT DETAILS -----

Savings Account
account holder nameNaveen
account no is 101
balance in account9550
account typesavings
```

Current Account
account holder nameRahul
balance in account-2000
account typecurrent

```
PS C:\Users\navee\OneDrive\Desktop\programs> javac TaskII.java
PS C:\Users\navee\OneDrive\Desktop\programs> java TaskII
----- SAVINGS ACCOUNT DETAILS -----
account holder nameNaveen
account no is 101
balance in account10000
account typesavings

----- CURRENT ACCOUNT DETAILS -----
account holder nameRahul
account no is 102
balance in account5000
account typecurrent

----- TRANSACTIONS -----
amount is deposited into account succesfully : 2000
balance updated to 12000
withdraw suscessfully :1000
balance is upated to :11000
interest is 550.0
balance after interest is 11550
amount is deposited into account succesfully : 2000
balance updated to 7000
withdraw suscessfully :9000
balance is upated to :-2000
transacation done successfully
balance in acc is :9550
```

```
----- FINAL ACCOUNT DETAILS -----

Savings Account
account holder nameNaveen
account no is 101
balance in account9550
account typesavings

Current Account
account holder nameRahul
account no is 102
balance in account0
account typecurrent
```